ZCAS UNIVERSITY

CIT2351 NETWORK DESIGN AND SECURITY

FINAL EXAMINATIONS

TUESDAY, 26TH MAY 2026

14:00-17:00 HOURS

3hours

TIME ALLOWED: ~~ONE AND HALF HOURS~~ (plus 5 minutes to read through the paper)

INSTRUCTIONS TO CANDIDATES:

1. **SECTION A:** This section is compulsory. You must answer **Question 1.**
2. **SECTION B:** Attempt ~~**any three (03)** questions out of the **four (04)**~~ questions provided. ALL
3. Time allowed: **3 Hours plus five Minutes to read through the paper.**
4. This examination paper carries a total of **100 marks.**
5. Candidates **MUST NOT** turn this page **until** the Invigilator tells them to do so.

SECTION A: This section is compulsory. You must answer Question ONE.

QUESTION ONE

SCENARIO

Best Logistics is a Lusaka based logistics company with two offices: one in Longacres and the other one in Marshlands. The connection between two offices consists of two routers (**R1** and **R2**) which are connected via a serial link. Each router connects to a local LAN:

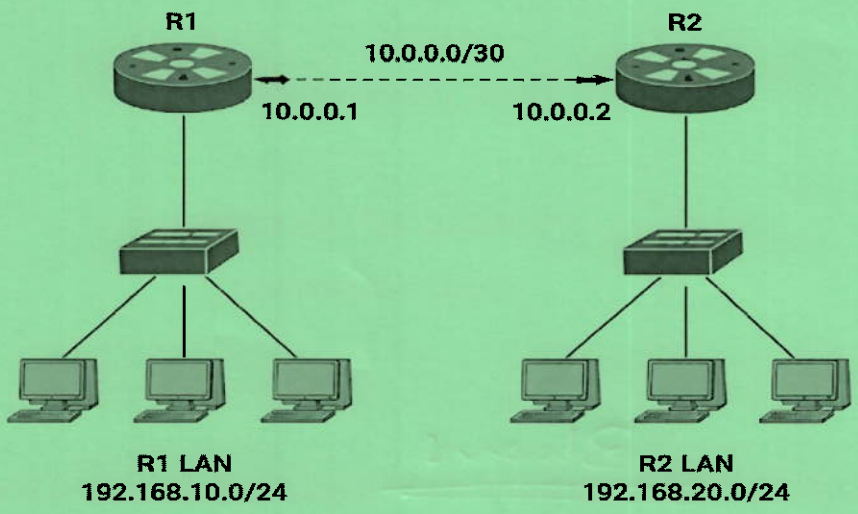


Figure 1: Mondi Logistics network

[R1 LAN: 192.168.10.0/24], [R2 LAN: 192.168.20.0/24], [Serial Link: 10.0.0.0/30]

You are required to configure routing and connectivity between the networks.

REQUIRED:

a) Write the basic configuration commands to assign IP addresses to all interfaces on both routers. **(10 marks)**

b) Configure RIPv2 on both routers to enable communication between the two LANs. **(10 marks)**

c) Reconfigure the network to use OSPF (Process ID 1, Area 0) instead of RIP. **(10marks)**

d) Configure PPP encapsulation on the serial link between R1 and R2. **(05 marks)**

e) Provide verification commands to confirm that routing is working correctly and explain the expected outputs. **(05 marks)**

(TOTAL 40 MARKS)

QUESTION TWO

SCENARIO

A fast-growing Zambian logistics company, ***ZamLogistics Ltd.***, is expanding operations across multiple provinces. The company currently uses a small, poorly structured network that frequently experiences downtime and slow performance. You have been hired as a network consultant to redesign their network using a top-down design approach.

REQUIRED:

a) Identify and explain two fundamental business goals that should guide the new network design. **(05 marks)**

b) Using the top-down network design approach, explain the first two steps you would take when engaging ***ZamLogistics Ltd.*** **(05 marks)**

c) Describe how the PDIOO lifecycle would apply to this project. **(05 marks)**

d) Identify two business constraints that could affect the network design and explain their impact. **(05 marks)**

(Total: 20 marks)

QUESTION THREE

SCENARIO

The National Revenue Authority plans to interconnect its regional offices and enable secure access to its tax processing system. To achieve this, they have sought your expert advice on the network design and security aspects.

REQUIRED:

a) Explain how the use of Dynamic Host Configuration Protocol (DHCP) simplifies IP address management in large networks. **(05 marks)**

b) What is the importance of including network logs in the overall security monitoring strategy? **(05 marks)**

c) Differentiate between wired and wireless networks in terms of security vulnerabilities and how to mitigate them. **(05 marks)**

d) Explain the purpose of including a firewall in network design and compare IDS with IPS in terms of function and deployment. **(05 marks)**

(Total: 20 marks)

QUESTION FOUR

SCENARIO

A new ZCAS University Kabelenga campus is being built and requires a robust, scalable, and secure network infrastructure to support thousands of users, including staff and students. The network must also support reliable wireless access across the entire campus.

REQUIRED:

You are hired as a Network Engineer for ZCAS University, and you are tasked with implementing security policies to protect sensitive student data.

a) Explain the importance of a Demilitarized Zone (DMZ) in securing externally-facing services such as the university's web servers. **(05 marks)**

b) Compare and contrast Layer 2 and Layer 3 network devices in the context of segmentation and traffic control. **(05 marks)**

c) Explain the importance of network security protocols in network design and identify TWO (02) security measures you would implement to protect this network. **(05 marks)**

d) During your network management, you have detected a network security breach. Describe the steps you would take to respond to this incident and mitigate its impact. **(05 marks)**

(Total: 20 marks)

END OF EXAMINATIONS